18. Limitations and Risk Register

Analytical limitations

1. **Unknown provenance:** no source URL, collection method, license, or sampling design was included.
2. **Representativeness:** counts by category, manufacturer, country, or region are not automatically market share.
3. **Unspecified definitions:** cycle life, CO■ saved, annual sales, autonomy level, charging time, and customer rating need source documentation.
4. **Missingness:** 9,511 cells are missing, including 2,028 cycle-life values and 1,931 customer ratings.
5. **Outliers:** maximum price is $409,995.49; maximum annual sales is 772,442; maximum efficiency is 2,242.84 Wh/km. Verify these records before interpretation.
6. **Correlation:** relationships are descriptive and do not establish causal effects.
7. **Time coverage:** model years run from 2018 to 2026, but record counts by year should not be interpreted as industry growth.
8. **Environmental estimates:** CO■ savings require a baseline and assumptions not included in the file.

Risk register

ID | Risk | Likelihood | Impact | Mitigation | Status

R-01 | Unknown data provenance | High | High | Seek source documentation; disclose uncertainty | Open

R-02 | Ambiguous sales field | Medium | High | Verify scope and avoid unsupported market-share claims | Open

R-03 | Extreme values distort averages | High | Medium | Use median/percentiles and inspect records | Open

R-04 | Missing metric values | Medium | Medium | Report denominator and use metric-specific null handling | Managed

R-05 | Embed blocked or slow | Medium | Medium | Public permissions and direct links as fallback | Pending live test

R-06 | Overstated environmental conclusions | Medium | High | Document CO■ calculation basis or treat as supplied estimate | Open

Future work

Obtain source documentation; validate a sample of records; define calculation bases; compare with trusted external datasets if permitted; and, only after data validation, consider segmentation or modeling.